

TENZIN THINLAY BHUTIA

Full Stack Developer • RAG Systems • LLM Integration • AI

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PROFESSIONAL SUMMARY

Full Stack Developer with 4+ years of production experience in React, Next.js, TypeScript, Python, and FastAPI — specializing in GenAI and LLM-powered applications. Built and deployed AskMyDocs — a full stack RAG SaaS with PDF ingestion, Gemini embeddings, pgvector semantic search, and JWT auth, deployed on Vercel + AWS EC2. Also built DeepTrail — a production RAG system live on the Chrome Web Store. Combines deep engineering fundamentals (Docker, CI/CD, OAuth, Redis, AWS S3) with hands-on GenAI work: RAG architectures, embeddings, prompt engineering, agentic systems, and vector search.

TECHNICAL SKILLS

Frontend:	React, Next.js 15 (App Router), TypeScript, JavaScript, Tailwind CSS, shadcn/ui, Chrome Extensions (Manifest V3), React Flow
Backend:	Python, FastAPI, Node.js, Express, PostgreSQL, pgvector, MongoDB, Redis, asyncpg
AI / GenAI:	RAG Pipelines, Vector Search, Gemini Embeddings, Prompt Engineering, LLM APIs (Gemini, Groq, OpenAI), Agentic Systems (ReAct, Tool Use), pgvector, Semantic Search, Evaluation (RAGAS), LangChain, MCP (Model Context Protocol)
Infrastructure:	Docker, AWS EC2, AWS S3, GitLab CI/CD, GitHub Actions, Vercel, Neon.tech, Render, New Relic, Harbor
Auth & Security:	JWT (httpOnly cookies), OAuth 2.0, NextAuth, WSO2, ForgeRock, bcrypt, Redis-based locking
AI Dev Tools:	Claude Code, Cursor, GitHub Copilot, v0, Prompt-Driven Development

PROJECTS

AskMyDocs — Full Stack RAG Document Q&A SaaS Apr 2026 – Present

Stack: Next.js 15, FastAPI, PostgreSQL + pgvector, Gemini Embeddings, AWS S3, AWS EC2, Vercel, Neon.tech, JWT, Docker
GitHub | Live Demo

- Built and deployed a production-grade RAG SaaS with PDF ingestion, semantic chunking, Gemini embeddings, pgvector retrieval, and context-grounded LLM response synthesis.
- Architected a secure Next.js 15 App Router frontend with httpOnly JWT cookies, server components, middleware-based route protection, and Next.js API routes as a proxy layer to FastAPI.
- Implemented AWS S3 for scalable PDF storage with automatic cleanup on document deletion, and asyncpg connection pooling for high-performance PostgreSQL queries.
- Built a full authentication system — register, login, logout, and forgot/reset password with time-limited tokens and Resend email delivery.
- Deployed backend on AWS EC2 with Docker, frontend on Vercel, and database on Neon.tech — fully production-ready stack with CORS, environment-based config, and restart policies.

DeepTrail — AI-Powered Research Knowledge Graph Mar 2026 – Present

Stack: Chrome Extension (MV3 + React 18), FastAPI, PostgreSQL 16 + pgvector, Gemini Embeddings (768-dim), Groq / Llama 3, React Flow, Neon.tech, Render
GitHub | Chrome Web Store

- Designed and built a production RAG pipeline: content extraction from browser activity → Gemini embedding (768-dim) → pgvector cosine similarity search → Groq LLM synthesis with context-grounded responses.

- Chose pgvector over Pinecone for hybrid query support (vector similarity + SQL metadata filtering in single queries) and ACID transactions ensuring atomic embedding + metadata writes.
- Built interactive knowledge graph visualization with React Flow (D3-force cluster layout) and implemented cosine similarity-based duplicate detection to maintain graph quality.
- Designing freemium SaaS architecture (Free / Pro \$8/mo / Team \$20/mo) with Stripe integration, model routing for cost optimization, and semantic caching for repeated queries.

Lumen AI — LLM-Powered Browser Assistant

Oct 2025 – Present

Stack: JavaScript, Chrome Extension (Manifest V3), Groq Cloud API, Llama 3.1, Marked.js

GitHub

- Built a Chrome Extension providing real-time AI analysis of web content via Groq's Llama 3.1 inference, with context-aware summarization and one-click AI explanations via Context Menus API.
- Implemented Background Service Worker for async API orchestration and BYOK vault using chrome.storage for zero-knowledge privacy.

PROFESSIONAL EXPERIENCE

Senior Developer

Dec 2021 – Present

Tecnotree Convergence Pvt Ltd — Bangalore, India

- Developed LLM-powered retrieval workflows and AI tooling integrations using FastAPI, vector search, and modern GenAI application patterns alongside enterprise platform engineering.
- Integrated MCP (Model Context Protocol) server workflows and TypeScript-based AI tooling over TMF 671 APIs with token pass-through authentication and structured tool orchestration.
- Designed asynchronous data-processing and transformation pipelines with validation, retry handling, and distributed workflow coordination — architectural patterns later applied in RAG ingestion and embedding systems.
- Integrated New Relic Browser Agent with structured logging (Pino) for observability, monitoring, and debugging across distributed Next.js services and API workflows.
- Architected OAuth 2.0 SSO authentication using WSO2, ForgeRock, and NextAuth; resolved distributed token refresh race conditions using Redis-based synchronization and locking mechanisms.
- Managed Dockerized deployments, Harbor image security, and GitLab CI/CD automation for enterprise applications; improved deployment reliability and reduced production vulnerabilities.
- Improved frontend performance by 40% through code-splitting, lazy loading, caching optimization, and efficient rendering strategies across large-scale React/Next.js applications.
- Leveraged AI-assisted engineering workflows using Claude Code, Cursor, GitHub Copilot, and prompt-driven development practices to accelerate feature delivery and developer productivity.

EDUCATION

B.Tech — Electronics & Communication Engineering

2017 – 2021

Surendra Institute of Engineering & Management | GPA: 3.7 / 4.0

LANGUAGES

English (Fluent), Hindi (Fluent), Nepali (Native), Tibetan (Native)